

- 6 A skateboarder of mass m skates on a smooth semi-circular half-pipe as shown. He starts from point X from rest and skates towards Y. Assume that air resistance is negligible.



What is the maximum force exerted by the half-pipe on the skateboarder?

- A mg
- B $2mg$
- C $3mg$
- D $4mg$

- 7 The diagrams shows three equipotential lines around a point mass. The dotted lines